Cost

- Cost is the *economic value of goods and services that producers and consumers purchase*
- Project *cost estimation* is the process of *predicting* the *quantity, cost,* and *price* of the *resources* required by the scope of a project.
  - Cost estimation is about the *prediction of costs rather than counting the actual cost*.
  - A certain *degree of uncertainty is involved because project scope definition is never entirely completed* until the project has been finished, and at this point all expenses have been made and an accountant can determine the exact amount of money spent on resources.
- There are a *variety of costs* to be considered in an engineering economic analysis.

Variable Costs

- Variable costs are costs that *change* as the *quantity of the good or service* that a business produces *changes*.
- *A company's variable cost increases and decreases with its production volume*. When production volume goes up, the variable costs will increase. On the other hand, if the volume goes down, so too will the variable costs.
- Variable costs can be calculated by multiplying the quantity of output by the variable cost per unit of output.

Variable Costs

Example

- Suppose a company ABC produces ceramic mugs for a cost of \$2 a mug. If the company produces 500 units (mugs), its variable cost will be 500 x 2 = \$1000.
- Other examples of variable costs include labour costs, utility costs, commissions, and the cost of raw materials that are used in production.

Fixed Costs

- Unlike the variable cost, *a company's fixed cost does not vary with the volume of production*. It remains the same even if no goods or services are produced, and therefore cannot be avoided.
- Using the same previous example, suppose company ABC has a fixed cost of $10,000 per month for the rent of the machine it used to produce mugs. If the company does not produce any mugs for the month, it would still have to pay $10,000 for the cost of renting the machine.

Recurring Cost

They are *repetitive* and *occur when a firm produces similar goods* and *services on a continuing basis.*

- Variable costs are recurring because they repeat with each unit of output.
- A fixed cost that is paid on a repeatable basis is also a recurring cost
  - Office space rental
  - Weekly maintenance costs

Non-Recurring

- *Unusual charge, expense, or loss that is unlikely to occur again in the normal course of a business*.
  - Non recurring costs include write offs such as design, development, and investment costs, and fire or theft losses, lawsuit payments, losses on sale of assets, and moving expenses.

Life-Cycle Cost

- *Summation of all the costs* related to a product, structure, system, or service during its life span.
- The *life cycle begins with identification* of the requirements and *ends with retirement and disposal activities*.

Other Costs

- ***Investment Cost:*** Money required for most activities of the acquisition phase.
- ***Operation and Maintenance:*** Cost includes many of the recurring annual expense items associated with the operation phase of the life cycle.
- ***Disposal Cost:*** includes non-recurring costs of shutting down the operation.

Cost-Volume Analysis

- Cost-volume analysis focuses on relationships between ***cost***, ***revenue***, and ***volume of the output***. The purpose of cost-volume analysis is ***to estimate the income of an organization under different operating conditions***. It is particularly useful as a tool for comparing alternatives.
- Use of the technique requires identification of all costs related to the production of a given product. These costs are often designated as fixed costs or variable costs. Fixed costs tend to remain constant regardless of volume of output. Variable costs vary directly with volume of output.
- ***We will assume that variable cost per unit remains the same regardless of volume of output, and that all output can be sold***.
- Table 1 next slide summarizes the symbols used in the cost-volume formulas.

Cost-Volume Analysis

- The total cost (TC) associated with a given volume of output is equal to the sum of the fixed cost (FC) and the variable cost (VC) per unit times volume:

$$\text{TC} = \text{FC} + \text{VC}$$

$$\text{VC} = Q \times v$$

- Figure A next slide shows the relationship between volume of output (Q), Fixed Cost (FC), Total Variable Cost (VC) and Total Cost (TC).

Cost–volume symbols

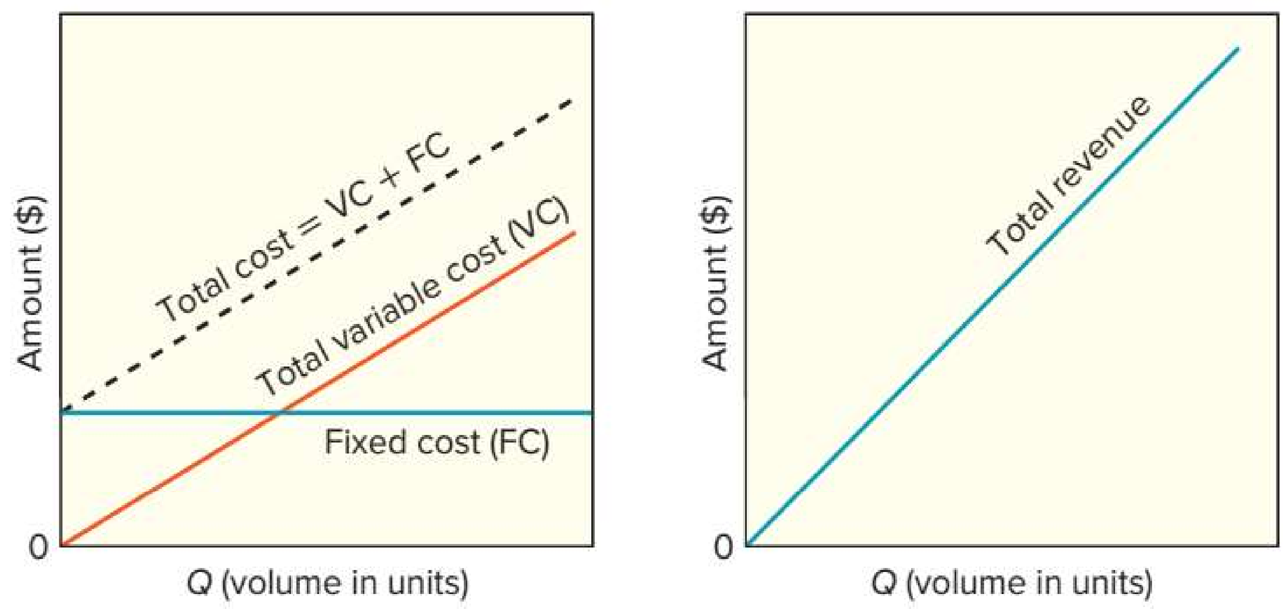
FC = Fixed cost
VC = Total variable cost
v = Variable cost per unit
TC = Total cost
TR = Total revenue
R = Revenue per unit
Q = Quantity or volume of output
Q_{BEP} = Break-even quantity
P = Profit

Table 1

Cost-Volume Analysis

- Revenue per unit, like variable cost per unit, is assumed to be the same regardless of quantity of output. Total revenue (TR) will have a linear relationship to output (*Q*) shown below in figure B. The total revenue (TR) associated with a given quantity of output, *Q*, is:
  TR= *R* x *Q*

Figure A and B



A. Fixed, variable, and total costs

B. Total revenue increases linearly with output

Cost-Volume Analysis

- Figure C shown below describes the ***profit***- which is the difference between total revenue (TR) and total (i.e., fixed plus variable) cost (TC). The volume at which total cost and total revenue are equal is referred to as the ***break-even point (BEP)***.

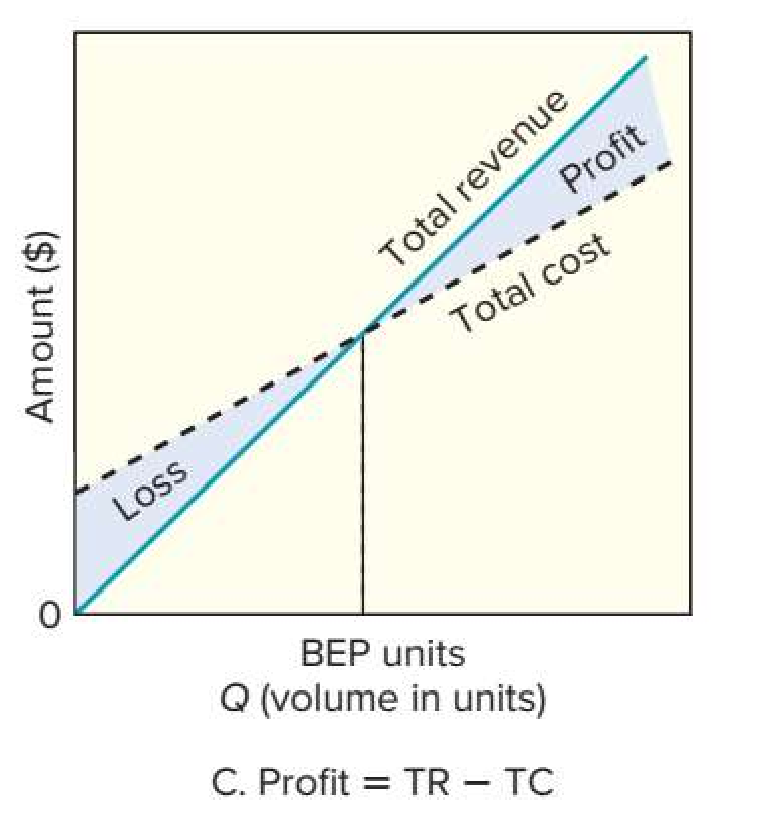


Figure C

Cost-Volume Analysis

- When volume is less than the break-even point, there is a loss; when volume is greater than the break-even point, there is a profit. The greater the deviation from BEP point, the greater the profit or loss. Figure D shown below describes total profit or loss relative to the BEP. Figure D can be obtained from figure C by drawing a horizontal line through the point where the total cost and total revenue lines intersect.

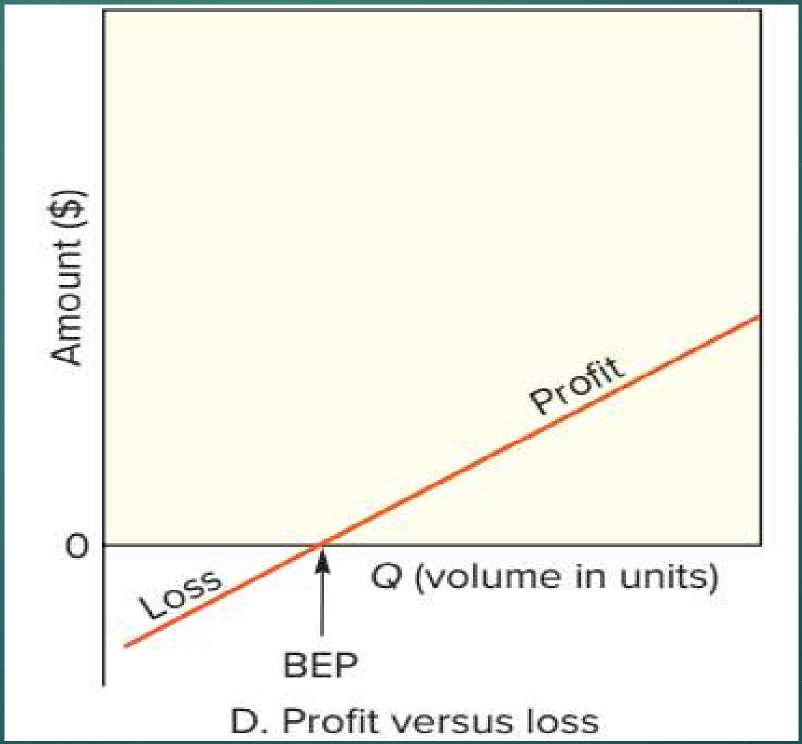


Figure D

Cost-Volume Analysis

- Total profit can be computed using the formula

$$P = \text{TR} - \text{TC} = R \times Q - (\text{FC} + v \times Q)$$

Rearranging terms, we have

- ***If Profit = +ve; it's indeed a profit***
- ***If Profit = -ve; it's indeed a loss.***

$$P = Q(R - v) - \text{FC}$$

The difference between revenue per unit and variable cost per unit, $R - v$, is known as the *contribution margin.*

The required volume, Q, needed to generate a specified profit is

$$Q = \frac{P + \text{FC}}{R - v}$$

A special case of this is the volume of output needed for total revenue to equal total cost. This is the break-even point, computed using the formula

$$Q_{\text{BEP}} = \frac{\text{FC}}{R - v}$$

Cost-Volume Analysis

Example:

Break-even Point and Profit Analysis

The owner of Old-Fashioned Berry Pies, S. Simon, is contemplating adding a new line of pies, which will require leasing new equipment for a monthly payment of $6,000. Variable costs would be $2 per pie, and pies would retail for $7 each.

a. How many pies must be sold in order to break even?

b. What would the profit/**loss** be if 1,000 pies are made and sold in a month?

c. How many pies must be sold to realize a profit of $4,000?

d. If 2,000 can be sold, and a profit target is $5,000, what price should be charged per pie?

Cost-Volume Analysis

Solution:

FC = \$6,000, VC = \$2 per pie, R = \$7 per pie

a. $$Q_{\text{BEP}} = \frac{\text{FC}}{\text{R} - \text{VC}} = \frac{\$6{,}000}{\$7 - \$2} = 1{,}200 \text{ pies/month}$$

b. For $Q = 1{,}000$, $P = Q(R - v) - \text{FC} = 1{,}000\ (\$7 - \$2) - \$6{,}000 = -\$1{,}000$

c. $P = \$4{,}000$; solve for Q

$$Q = \frac{\$4{,}000 + \$6{,}000}{\$7 - \$2} = 2{,}000 \text{ pies}$$

d. $\text{Profit} = Q(R - v) - \text{FC}$

$$\$5{,}000 = 2{,}000(R - \$2) - \$6{,}000$$

$$R = \$7.50$$

Reference(s)

- *Based in part on materials by Ms. Anam Rafique (Assistant Professor, Department of Computer Engineering in HITEC University Taxila Cantt).*
- *Lucy C.Morse & Daniel L.Babcock, "Managing Engineering and Technology", 6th Edition, Pearson.*
- *John M.Nicholas & Herman Steyn, "Project Management for Business, Engineering and Technology, Principles and practice", 3th Edition, ELSEVIER.*
- *Harold Kerzner, "Project Management, A System Approach to Planning, Scheduling & Controlling", 10th Edition, John Wiley & Sons, Inc, 2012, ISBN: 978-0-470-27870-3.*
- *Kathryn Bartol, David Martin, Margaret Tein, Graham Matthews "Management, A Pacific Rim Focus" 3rd Edition, McGraw-Hill Companies, Inc, 2002.*
- *Gerry Johnson & kevan Scholes, "Exploring Corporate Strategy", 8th Edition.*
- *Related documents from open source, mainly internet.*